

Problem Set 5
Due: Mar. 29, 2021

1. Section 2.6 Problem 32.

D. Bernoulli's problem of joint lives. Consider $2m$ persons forming m couples who live together at a given time. Suppose that at some later time, the probability of each person being alive is p , independent of other persons. At that later time, let A be the number of persons that are alive and let S be the number of couples in which both partners are alive. For any survivor number a , find $\mathbf{E}[S \mid A = a]$.

2. Section 2.7 Problem 42.

Computational problem. Here is a probabilistic method for computing the area of a given subset S of the unit square. The method uses a sequence of independent random selections of points in the unit square $[0, 1] \times [0, 1]$, according to a uniform probability law. If the i th point belongs to the subset S the value of a random variable X_i is set to 1, and otherwise it is set to 0. Let $X_1, X_2, \dots$ be the sequence of random variables thus defined, and for any n , let

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

- (a) Show that $\mathbf{E}[S_n]$ is equal to the area of the subset S , and that $\text{var}(S_n)$ diminishes to 0 as n increases.
 - (b) Show that to calculate S_n , it is sufficient to know S_{n-1} and X_n , so the past values of X_k , $k = 1, \dots, n-1$, do not need to be remembered. Give a formula.
 - (c) Write a computer program to generate S_n for $n = 1, 2, \dots, 10000$, using the computer's random number generator, for the case where the subset S is the circle inscribed within the unit square. How can you use your program to measure experimentally the value of π ?
 - (d) Use a similar computer program to calculate approximately the area of the set of all (x, y) that lie within the unit square and satisfy $0 \leq \cos \pi x + \sin \pi y \leq 1$.
3. Every package of some intrinsically dull commodity includes a small and exciting plastic object. There are c different types of object, and each package is equally likely to contain any given type. You buy one package each day.
- (a) Find the mean number of days which elapse between the acquisitions of the j th new type of object and the $(j+1)$ th new type.
 - (b) Find the mean number of days which elapse before you have a full set of objects.
4. Each trial may result in any of t given outcomes, the i th outcome having probability p_i . Let N_i be the number of occurrences of the i th outcome in n independent trials.

(a) Show that

$$P(N_i = n_i \text{ for } 1 \leq i \leq t) = \frac{n!}{n_1!n_2!\cdots n_t!} p_1^{n_1} p_2^{n_2} \cdots p_t^{n_t}$$

for any collection of $n_1, n_2, \dots, n_t$ of non-negative integers with sum n . The vector N is said to have the multinomial distribution.

(b) Find the marginal mass functions of the multinomial distribution.